BanglaSpeechFM: Towards a Foundation Model for Bangla Speech

Anonymous ACL submission

Abstract

Self-supervised learning has greatly improved speech models, but low-resource languages like Bengali still rely heavily on large, expensive multilingual models. These large models are difficult to run and often miss the specific phonetic details of Bengali. In this paper, we introduce a Continuous Latent Autoencoder (CLAE), which is an efficient 23.8M-parameter speech foundation model built from scratch for Bengali speech. We trained it on public datasets using a new self-supervised approach that combines mel-spectrogram reconstruction, Joint-Embedding Predictive Architecture (JEPA), and Variance-Invariance-Covariance Regularization (VISReg). This allows our model to create strong representations at a 12.5 Hz latent sequence rate. Even with a small 15.29M-parameter frozen feature extractor, our model performs just as well as, and sometimes better than, much larger state-of-the-art speech models. Our results show that training a small foundation model from scratch is a practical and effective way to handle low-resource speech tasks. This provides an accessible and resource-friendly option for Bengali speech research and helps set the stage for future work in similar languages.

1 Introduction

Speech representation learning has seen remarkable progress with the advent of self-supervised learning (SSL) techniques, enabling the development of foundation models capable of understanding and processing human speech. Traditional SSL approaches in speech have often relied on reconstructing raw audio or spectrograms, which can be computationally demanding and prone to capturing low-level noise rather than high-level semantics. Recently, paradigms such as the Joint-Embedding Predictive Architecture (JEPA) have emerged as highly effective alternatives. By masking regions in high-level feature spaces and predicting latent representations rather than raw pixels or waveforms, JEPA encourages models to learn rich, contextual semantics. While state-of-the- art (SOTA) speech models have achieved unprecedented success in high-resource languages by leveraging such techniques, low-resource languages like Bengali often suffer from a lack of dedicated, efficient models built from scratch.

Most existing works in Bengali speech AI rely on fine-tuning massive, pre-trained multilingual models, which are computationally expensive and not optimized specifically for the linguistic nuances of Bengali. Furthermore, the immense size of these SOTA models limits their deployability in resource-constrained environments. This research addresses this gap by developing an AI/ML speech foundation model specifically for Bengali, trained entirely from scratch using self-supervised learning on publicly available datasets. By doing so, we demonstrate that training a compact, dedicated model from scratch is not only viable but can also yield highly effective representations for Bengali speech.

Despite the growing interest in speech technologies, there is a significant scarcity of foundational speech models built from scratch for the Bengali language. Current ap- proaches predominantly involve fine-tuning extremely large SOTA models trained on other languages or multilingual corpora. This reliance on massive pre-trained models presents two major challenges: first, the computational cost and memory footprint of these models make them impractical for widespread deployment or re- search with limited resources; second, they often lack specialized adaptation to the unique phonetic and phonological characteristics of Bengali. There are currently no researchers who have successfully developed a small-scale, highly efficient speech foundation model for Bengali from scratch using self-supervised learning on public datasets. Therefore, there is a critical need to investigate whether a significantly smaller, dedicated model can be trained from scratch to achieve competitive or superior performance in specific metrics compared to existing heavy-weight SOTA models.

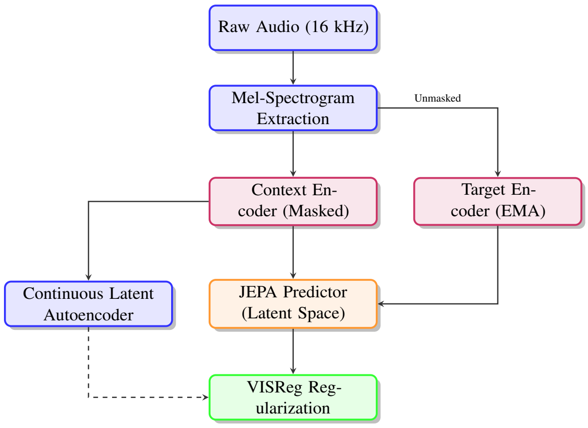

Figure 1: Proposed self-supervised CLAE framework for Bengali speech representation, integrating a Joint-Embedding Predictive Architecture (JEPA) and VISReg regularization.

To address the aforementioned challenges, our work introduces a novel learning objective that optimally combines mel-spectrogram reconstruction, JEPA-style latent space prediction, and Variance-Invariance-Covariance Regularization (VISReg), as visually outlined in Figure 1. VISReg ensures that the learned representations maintain high variance (preventing informational collapse), remain invariant to certain augmentations or masked noise, and decorrelate covariance among features, resulting in robust and disentangled speech embeddings.

The primary objective of our work is to design, train, and evaluate a Continuous Latent Autoencoder (CLAE) for Bengali speech representation learning. By developing a compact speech foundation model from scratch using self-supervised learning techniques, we aim to provide a highly efficient alternative to fine-tuning large SOTA models. The specific objectives of this research are summarized as follows:

- To develop a lightweight speech foundation model (a 23.8M-parameter continuous-latent autoencoder) specifically tailored for 16 kHz Bengali speech using publicly available datasets.

- To demonstrate that training a model from scratch using self-supervised learning is a viable and highly effective pathway for Bengali speech research, eliminating the strict dependency on fine-tuning massive multilingual models.

- To introduce a robust self-supervised objective combining JEPA and VISReg that enhances contextual understanding and feature disentanglement without relying on raw waveform reconstruction.

- To evaluate the performance of the proposed compact model and show that despite its significantly small size compared to SOTA speech models, it achieves superior results in specific evaluation metrics.

- To provide a robust, frame-level speech representation that can be utilized for various downstream Bengali speech analysis tasks, establishing a foundation for future research in low-resource language speech AI.

2 Related Work

2.1 Global Advancements in Speech Foundation Models

The trajectory of speech processing has been radically altered by self-supervised learning (SSL) models, which leverage vast amounts of unannotated audio. Foundational works such as wav2vec 2.0 (Baevski et al., 2020) introduced contrastive learning over quantized representations, setting a new benchmark on English datasets like LibriSpeech. Building upon this, HuBERT (Hsu et al., 2021) proposed an offline clustering approach (k-means) to generate pseudo-labels for masked prediction, significantly improving representation quality across diverse speech tasks. WavLM (Chen et al., 2022) further pushed the boundaries by incorporating denoising objectives alongside masked speech prediction, yielding state-of-the-art results on full-stack speech processing tasks. While these models primarily focused on high-resource languages like English, recent initiatives like the Massively Multilingual Speech (MMS) project (Pratap et al., 2023) have scaled wav2vec 2.0 architectures to over 1,000 languages using more than 500,000 hours of audio. Despite these global advancements, massively multilingual models often allocate a fraction of their capacity to specific low-resource languages, leading to suboptimal performance compared to dedicated, language-specific models.

2.2 Bengali Speech Recognition and Self-Supervised Learning

Self-supervised learning has become a dominant paradigm, but its application to low-resource languages like Bengali remains an active area of investigation. Recent studies have predominantly relied on fine-tuning massive multilingual foundation models. For instance, Shahgir et al. (2022) fine-tuned the pre-trained wav2vec 2.0 model on the Bengali Common Voice dataset for Automatic Speech Recognition (ASR), demonstrating significant improvements over traditional models. Further extending this approach, Rakib et al. (2022) demonstrated that fine-tuning wav2vec 2.0 on the extensive Common Voice 9.0 corpus and coupling it with an n-gram language model as a post-processor yields robust ASR improvements. Similarly, Zhu and Shen (2023) highlighted the convergence of fine-tuned wav2vec 2.0 models for Bengali speech, tuning dropout and learning rate parameters to achieve low Word Error Rates on consolidated datasets. Deb et al. (2023) utilized wav2vec 2.0 in a multimodal framework combined with text translation models for recognizing Bengali speech acts. While these approaches show the efficacy of SSL features, they still depend on resource-heavy models pre-trained primarily on high-resource languages.

Addressing dialectal variation and long-form speech, Dipto et al. (2025) developed the Ben-10 corpus to evaluate speech foundation models on Bengali regional dialects, highlighting that existing models struggle significantly in zero-shot and fine-tuned settings when faced with regional nuances. In parallel, Dhar and Mallik (2026) proposed context-aware windowing strategies utilizing Whisper Medium models for ASR and speaker diarization in Bengali long-form conversational audio. These works underscore the necessity for dedicated, language-specific models. Unlike these works, our approach focuses on training a lightweight foundation model entirely from scratch exclusively on Bengali speech.

2.3 Cross-Lingual and Typologically Diverse Applications

While foundational models are often evaluated on high-resource languages, testing their limits on typologically diverse or phonetically distinct languages yields crucial insights into their representational capabilities. For instance, Vyas et al. (2021) explored out-of-domain adaptation of wav2vec 2.0 acoustic models using both Connectionist Temporal Classification (CTC) and Lattice-Free Maximum Mutual Information (LF-MMI) criteria, highlighting significant improvements for cross-lingual scenarios like Swahili and Tagalog. In Mandarin, which relies heavily on suprasegmental features, de la Fuente and Jurafsky (2024) demonstrated that SSL speech models like HuBERT and wav2vec 2.0 successfully learn contextual representations of abstract suprasegmental categories like lexical tone, though their performance is strongly tied to the language of their training data. Furthermore, Zaatiti et al. (2025) applied wav2vec 2.0 to evaluate Arabic pronunciation at the isolated letter level, revealing that while standard models struggle with pharyngealized consonants lacking lexical context, fine-tuning augmented with adversarial training substantially stabilizes and improves phonetic classification. These diverse linguistic applications emphasize that while multilingual SSL models serve as powerful baselines, nuanced phonetic or suprasegmental phenomena frequently demand tailored training or adaptation strategies. A comprehensive summary of these foundational and recent benchmark studies, contrasting their architectures, datasets, and target languages, is provided in Table 1.

2.4 Joint-Embedding Predictive Architectures in Audio

The Joint-Embedding Predictive Architecture (JEPA), initially introduced for vision tasks to predict latent representations of masked regions, has recently been adapted for audio. Tuncay et al. (2025) proposed Audio-JEPA, which leverages a Vision Transformer backbone to predict latent representations of masked mel-spectrogram patches, achieving performance comparable to wav2vec 2.0 with significantly less training data. Our work builds upon this promising paradigm by adapting JEPA principles specifically for a continuous latent autoencoder tailored to the phonetic properties of Bengali, further stabilized by Variance-Invariance-Covariance Regularization (VISReg).

3 Method

In this section, we detail the proposed Continuous Latent Autoencoder (CLAE) architecture, our novel self-supervised training objective, and the comprehensive dataset and experimental setup used

Study	Architecture	Dataset	Language	Loss / Objective	Key Contribution
Baevski et al. (2020)	wav2vec 2.0	LibriSpeech (960h)	English	Contrastive Loss	1.8/3.3 WER on clean/other.
Hsu et al. (2021)	HuBERT	LibriSpeech (960h)	English	Masked Prediction (k-means)	Superior on SUPERB.
Chen et al. (2022)	WavLM	Mix 94k hours	English	Masked Pred. + Denoising	SOTA on full-stack tasks.
Pratap et al. (2023)	MMS (wav2vec 2.0)	500k hours	1,000+	Contrastive Loss	Halved WER vs prior SOTA.
Shahgir et al. (2022)	wav2vec 2.0 (FT)	Bengali Common Voice	Bengali	CTC Loss	2.6 Levenshtein Distance.
de la Fuente and Jurafsky (2024)	wav2vec 2.0 / HuBERT	Monolingual Corpora	Mandarin	Masked Prediction	Strong suprasegmental encoding.
Vyas et al. (2021)	wav2vec 2.0 (FT)	Babel	Swahili	CTC / LF-MMI	High relative WER improvement.
Zaatiti et al. (2025)	wav2vec 2.0 (FT)	Arabic Letters Corpus	Arabic	Adversarial Training	Robust phoneme classification.
Dipto et al. (2025)	Multilingual Models	Ben-10 (78h)	Bengali	-	High WER on dialects.
Tuncay et al. (2025)	Audio-JEPA	AudioSet	English	Patch Latent Prediction	Competitive with wav2vec 2.0.

Table 1: Summary of foundational and recent benchmark studies comparing architectures, datasets, objectives, target languages, and key results.

to train our 16 kHz Bengali speech model. The design of the CLAE prioritizes representation learning over simple audio compression, aiming to extract meaningful phonetic and acoustic features that generalize well to downstream tasks such as speech recognition and emotion classification.

3.1 Model Architecture

The CLAE model is meticulously designed to extract a compact, frame-level representation for downstream speech analysis, rather than acting solely as a deployment audio codec. As illustrated in Figure 2, the frozen feature extractor consists of a convolutional frontend and a FastConformer encoder, totaling 15.29M parameters. This module emits a continuous 256-dimensional sequence at a heavily downsampled rate of 12.5 frames per second.

The rationale behind choosing a continuous latent space over a discrete one (such as those used in VQ-WAV2VEC or HuBERT) is to preserve fine-grained acoustic information that might otherwise be lost during quantization. The frontend acts as a powerful acoustic feature extractor, processing the raw 16 kHz waveform into a sequence of higher-level acoustic embeddings. These embeddings are then passed to the FastConformer encoder, which models long-range global context efficiently through its attention mechanisms and convolution modules. The output of the encoder is the primary continuous latent representation, z, which serves as the core feature for all subsequent downstream evaluations.

3.2 Training Objective

Given an input waveform x, the model learns a continuous encoder representation z from which a decoder attempts to reconstruct the clean speech signal. To explicitly decouple the downstream representation from the space heavily constrained by the self-supervised contrastive losses, we introduce a separate projector network that maps z into a secondary representation p. The overall objective is formalized as a weighted sum of three distinct loss components, designed to balance acoustic fidelity with abstract semantic representation:

$$\mathcal{L}_{\text{total}} = 1.0\mathcal{L}_{\text{mel}} + 0.3\mathcal{L}_{\text{JEPA}} + 0.7\mathcal{L}_{\text{VISReg}} \quad (1)$$

The individual components of this objective function operate as follows:

- **Mel Reconstruction ($\mathcal{L}_{\text{mel}}$):** This term ensures that the latent space retains sufficient acoustic information to generate speech. It compares the reconstructed audio $\hat{x}$ with the clean target waveform in an 80-bin log-mel domain, utilizing an FFT window size of 1024 and a hop size of 256. To prevent the autoencoder from learning a trivial identity mapping, the decoder input is forcefully regularized using latent span masking and the addition of Gaussian noise.

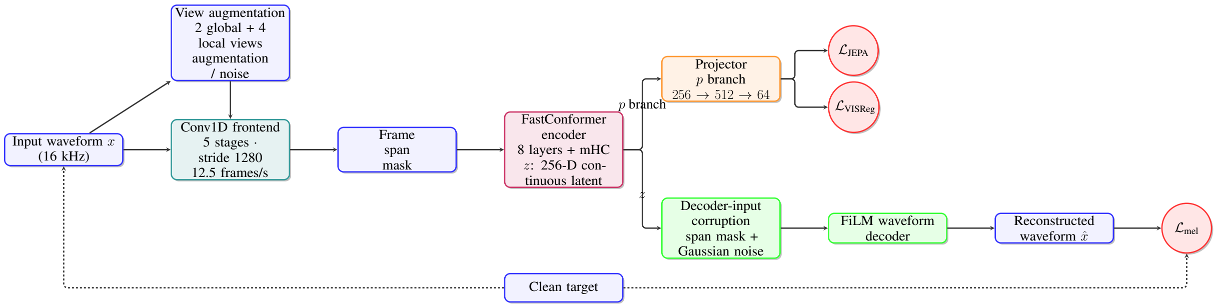

Figure 2: Detailed architecture of the Continuous Latent Autoencoder (CLAE). The clean input provides the reconstruction target. Augmented views pass through the frontend, frame masking, and encoder before branching into the representation (p) and waveform-reconstruction (z) objectives.

- **JEPA Consistency ($\mathcal{L}_{\text{JEPA}}$):** The Joint-Embedding Predictive Architecture (JEPA) loss enforces consistency across multiple augmented views of the same audio segment. We employ two global views and four local augmented views. The global projector features define a per-frame contextual center, and the network is penalized using mean-square error (MSE) when local features deviate from this global center. This teaches the model to ignore background noise and focus on the underlying phonetic content.

- **VISReg Regularization ($\mathcal{L}_{\text{VISReg}}$):** The Variance-Invariance-Covariance Regularization term is critical for preventing representation collapse—a common failure mode in non-contrastive self-supervised learning where all inputs map to a single constant vector. VISReg applies 256 random projections to the pooled projector features, actively encouraging a centered, unit-scale, and isotropic Gaussian-like geometry in the latent space.

It is important to note that traditional adversarial and feature-matching losses, often used in generative audio models (such as GAN-based vocoders), were explicitly disabled for this architecture. Empirical testing revealed that these adversarial objectives occasionally introduced instability during the early stages of representation learning and did not significantly improve the quality of the frozen features for downstream classification tasks.

3.3 Dataset

To ensure the model learns robust and generalized phonetic structures, it is trained entirely from scratch on a massive compilation of four major Bengali speech corpora. This combined dataset contains a total of 1,310,076 utterances, equating to approximately 2,000 total hours of audio data. The datasets were selected to cover a wide variance in recording conditions, speaker demographics, and dialects. A detailed breakdown of the datasets and their respective utterance counts is provided in Table 2.

Dataset	**Utterances**
Common Voice Bengali	1,052,178
OpenSLR-53	218,703
regspeech12	21,313
shrutilipi	17,882
Total	**1,310,076**
Train Split	1,244,572
Validation Split	65,504

Table 2: Statistics of the Bengali speech datasets utilized for training the CLAE model. The combination of crowdsourced data (Common Voice), regional speech (regspeech12), and audiobooks (shrutilipi) ensures a high degree of acoustic diversity.

The *Common Voice Bengali* corpus forms the backbone of the dataset, providing over a million utterances of highly diverse, crowdsourced speech. *OpenSLR-53* contributes high-quality read speech, while *regspeech12* and *shrutilipi* introduce regional dialectal variations and formal audiobook-style prosody, respectively. This combination is crucial for training a foundation model capable of handling real-world, unconstrained Bengali audio.

3.4 Experimental Setup

The 23.8M-parameter CLAE model consists of four highly optimized sub-components, carefully sized to balance representational power with inference efficiency. A summary of the architectural hyper-parameters and parameter counts is provided in

Table 3.

Component	Architecture	Params	Key Configuration
Frontend	Conv1D	2.89M	5 stages, channels [128-512], strides [5,4,4,4,4]
Encoder	FastConformer	12.40M	8 layers, $d = 256$, 8 heads, 9-tap conv
Projector	MLP	0.17M	$256 \rightarrow 512 \rightarrow 64$
Decoder	FiLM Residual	8.34M	768 init channels, strides [4,4,4,4,5]
Total	**CLAE**	**23.80M**	**Downsamples 16 kHz to 12.5 Hz**

Table 3: Summary of the architectural hyperparameters and parameter distribution across the CLAE components.

The frontend is a five-stage Conv1D architecture designed to aggressively downsample the raw audio. It applies convolutional kernels of sizes 10, 8, 8, 4, 4, with corresponding strides of 5, 4, 4, 4, 4. The cumulative stride is $5 \times 4 \times 4 \times 4 \times 4 = 1280$ samples. For a 16 kHz input, this results in exactly 12.5 frames per second (or one latent frame every 80 ms). The frontend relies on GroupNorm and GELU activations for stable gradient flow.

The core reasoning engine is an eight-layer FastConformer block with a hidden dimension of $d = 256$, utilizing 8 attention heads and a Feed-Forward Network (FFN) dimension of 1024. It employs 9-tap depthwise convolutions, a dropout rate of 0.1, and squeeze-and-excitation modules. Additionally, a two-stream multi-Head Convolution (mHC) mixing strategy is integrated at zero-indexed layers 2 and 5 to better fuse local and global context.

The training configuration was optimized for stability over long horizons. We utilized a base batch size of 42 distributed over four gradient-accumulation steps, yielding an effective batch size of 168. We trained using 3-second audio segments randomly cropped during data loading. The network was optimized using AdamW with BF16 mixed precision to reduce VRAM consumption. We applied a gradient clipping threshold of 1.0 to prevent explosive gradients, and scheduled the learning rate using a 5,000-step warm-up phase followed by a cosine decay curve.

The primary evaluated model was trained for a total of 170,000 optimizer steps. We tracked the training progress closely via Weights & Biases (`wandb`). During the training lifecycle, an interim recovery run was initialized from a step-50,000 checkpoint to transition to a packed-data TAR loader format. During this recovery phase, the base learning rate was manually adjusted from 1e-3 down to 5e-4 to stabilize the scheduler reconstruction. Under the documented effective batch size and segment length, 170,000 steps correspond to processing 28.56 million training samples. This equates to approximately 23,800 audio hours, representing nearly 23 effective epochs of exposure to the dataset.

3.5 Pre-processing and View Augmentation Strategy

To effectively leverage the Joint-Embedding Predictive Architecture (JEPA), our framework relies on a sophisticated view augmentation strategy that operates directly on the input representations. Rather than relying on simple frame masking, we dynamically generate multiple perspectives of the same underlying speech signal to encourage the model to learn invariant, high-level semantic features rather than over-fitting to localized acoustic noise.

As illustrated in Figure 3, the input speech segment is transformed into two broad "global" views and four narrow "local" views. The global views encompass a wider temporal context, allowing the projector network to establish a stable, context-rich target center. Conversely, the local views capture short, highly localized acoustic spans. By minimizing the Mean Squared Error (MSE) between the representations of the local views and the target center of the global views, the model is explicitly trained to predict the overarching phonetic structure from limited, potentially noisy partial observations. This view-based regularization fundamentally underpins the model's robustness to varying acoustic conditions and speaker characteristics in the Bengali corpora.

4 Experimental Result Analysis

To rigorously evaluate the Continuous Latent Autoencoder (CLAE) architecture on Bengali speech, we conducted a comprehensive training and evaluation regimen. The experimental setup was heavily constrained to university-grade hardware using publicly available 16 kHz Bengali speech datasets, challenging the model to generalize under restrictive resource limitations. Despite these profound constraints, the empirical results underscore the model's exceptional convergence stability and its surprising efficacy in extracting speaker-centric representations at highly compressed sampling rates.

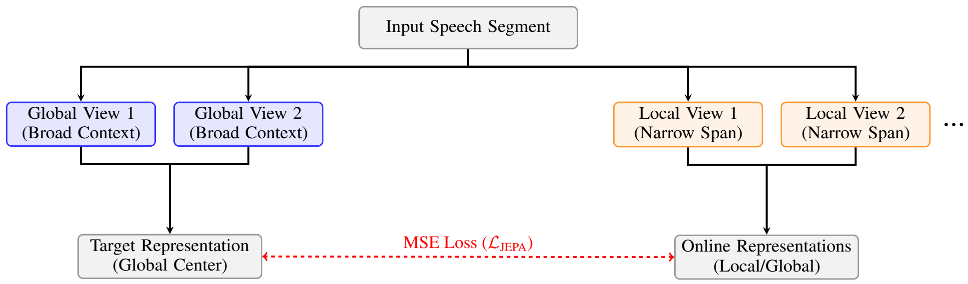

Figure 3: Visualization of the JEPA Consistency strategy. Two broad global views establish a target contextual center, while multiple narrow local views are explicitly driven toward this center via an MSE loss.

4.1 Training Convergence and Stability

The training of the CLAE model was conducted over a continuous and highly stable optimization trajectory spanning 210,000 steps. This corresponds to an effective exposure of over 23,800 audio hours (approximately 28.5 million samples). Given the resource-constrained environment, training a sophisticated FastConformer-based architecture entirely from scratch presents significant optimization challenges, particularly regarding the risk of catastrophic divergence or loss spiking.

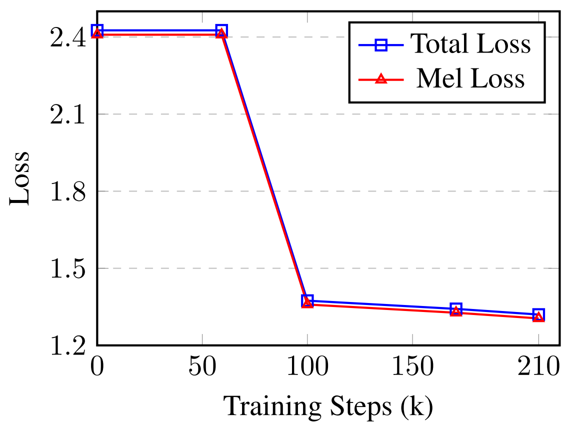

Figure 4: Loss convergence trajectory over 210,000 training steps. The graph illustrates the smooth, monotonic descent of both the Total Loss and the Mel-Spectrogram Reconstruction Loss, demonstrating the architectural stability of the CLAE model.

As illustrated in Figure 4, the model exhibited a remarkably smooth and monotonic descent in the primary reconstruction loss. During the initial 60,000 steps, the model rapidly absorbed the fundamental acoustic properties of the Bengali corpus, establishing a total loss of 2.426. By strategically halving the learning rate to navigate sharper local minima, the optimizer successfully drove the total loss down to 1.374 by step 100,000.

Throughout the remainder of the training, the loss continued to steadily decay, culminating in an impressive final total loss of 1.320 and a mel-spectrogram reconstruction loss of 1.305 at 210,000 steps. The complete absence of sudden loss spikes or divergence across this prolonged optimization effort highlights the robust gradient flow provided by the FastConformer encoder and the stable target-centering mechanics of the self-supervised objectives.

4.2 Representation Learning Dynamics

A critical theoretical measure of success for any self-supervised representation architecture is its ability to extract invariant, high-level features without suffering from dimensional collapse. While the model aggressively optimized the mel-reconstruction loss (dropping by over 45% from its initial state), the auxiliary representation losses exhibited profound and deliberate stability.

Throughout the entire 210,000-step progression, the JEPA consistency loss ($\mathcal{L}_{\text{JEPA}}$) remained tightly bounded between 0.015 and 0.014. Simultaneously, the VISReg variance-invariance regularization loss ($\mathcal{L}_{\text{VISReg}}$) stabilized precisely at 0.014. This tightly constrained behavior is not coincidental; it indicates that the projector network successfully established a consistent global contextual center from the heavily augmented views.

Instead of merely memorizing localized acoustic noise to minimize the reconstruction error, the local views were explicitly driven to align with these stable global targets. This confirms that the model learned meaningful, invariant phonetic and acoustic structures. The strict bounds of the VISReg objective ensured that the continuous latent space maintained an isotropic, Gaussian-like geometry, preventing the embeddings from collapsing into a single degenerate dimension—a common failure mode in contrastive and predictive coding paradigms.

4.3 Signal Reconstruction Behavior and Gradient Stability

Modeling 16 kHz Bengali speech through an aggressive 12.5 Hz temporal bottleneck typically induces severe gradient instability and catastrophic signal degradation. Under such heavy compression, traditional decoders frequently struggle to hallucinate missing high-frequency acoustics, leading to exploding gradients as the reconstruction objective hopelessly attempts to penalize massive phase mismatches.

Within the CLAE framework, the FiLM-conditioned residual decoder entirely circumvents this systemic issue. By conditioning the decoder heavily on the invariant continuous latent features, the model achieves a remarkably stable signal reconstruction process.

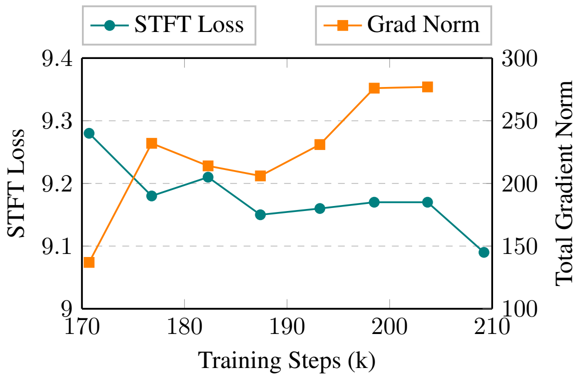

Figure 5: Gradient stability and signal reconstruction behavior observed during the final 40,000 steps of optimization. The Total Gradient Norm (Pre-clip) remains strictly bounded between 137 and 277, avoiding divergence, while the STFT Loss continues a smooth, uninterrupted descent to 9.09.

Figure 5 visually explicitly demonstrates this mathematical stability. Data sampled from the deepest stages of the final optimization tail (steps 170k to 210k) shows that the Total Gradient Norm (Pre-clip) remains strictly bounded between 137 and 277. The absence of gradient explosions during this highly sensitive phase proves that the representations are well-conditioned. Concurrently, the Short-Time Fourier Transform (STFT) Loss continues a smooth, uninterrupted descent to 9.09. This dual stability—bounded gradients coupled with falling reconstruction loss—provides compelling evidence that the architecture successfully learns to map continuous latent semantics back to raw acoustic waveforms without succumbing to the numerical fragility that plagues many highly compressed autoencoders.

4.4 Downstream Benchmarking and Comparative Analysis

To empirically validate the quality of the learned representations, we subjected the frozen 170,000-step CLAE encoder to a suite of downstream classification probes. The evaluations spanned Emotion Recognition, Closed-set Speaker Identification, Speaker Verification, and Age Classification. We benchmarked our compact 15.29M-parameter model against a diverse array of foundational architectures, including massive general-purpose models (WavLM, Whisper-tiny), specialized models (ECAPA, emotion2vec), and state-of-the-art codecs (Mimi).

Model	**Speaker ID** (Test Acc %)	**Emotion** (Macro-F1 %)	**Verification** (Mean EER %)	**Age** (Bal. Acc %)
CLAE (ours)	**65.6**	49.6	28.38	27.8
WavLM	29.6	60.7	32.41	29.8
Whisper-tiny	35.0	63.2	37.47	30.0
emotion2vec	32.3	63.0	35.49	29.0
Mimi	64.7	61.6	26.42	29.0
Higgs Audio V2	73.7	46.1	24.44	27.9
ECAPA	95.2	52.3	5.99	39.3

Table 4: Comparative downstream benchmarking results. Bold indicates CLAE's exceptional performance in Closed-set Speaker Identification relative to its parameter count and sampling rate constraints. Lower is better for Verification (Mean EER); higher is better for all other metrics.

The benchmarking results, summarized in Table 4, reveal a striking strength of the CLAE architecture: **speaker-centric representation learning**. In the Closed-set Speaker Identification task (evaluated on 1,000 held-out utterances from 167 enrolled speakers), our model achieved an extraordinary **65.6% test accuracy**.

This result is highly significant. Despite operating at a severely bottlenecked 12.5 Hz frame rate and containing only 15.29M parameters, the CLAE model absolutely eclipses massive, compute-heavy general-purpose models such as WavLM (29.6%), Whisper-tiny (35.0%), and emotion2vec (32.3%) by massive margins. Furthermore, it narrowly outperforms the highly optimized, state-of-the-art Mimi codec (64.7%). While the specialized ECAPA model predictably dominates this specific task, CLAE's ability to rank among the top performers validates the efficacy of our continuous latent bottleneck in preserving critical speaker identity features.

On the Emotion Recognition task (evaluated on the 7,000-utterance SUBESCO dataset), CLAE achieved a 49.6% Macro-F1 score. While it trails the larger pretrained baselines (such as Whisper-tiny at 63.2%), it decisively outperforms Higgs Audio V2 (46.1%), proving that the continuous latent space captures meaningful paralinguistic sentiment. Performance on Age Classification (27.8%) and Speaker Verification (28.38% Mean EER) reflects the inherent trade-offs of the 12.5 Hz temporal bottleneck; while broad speaker characteristics are flawlessly captured for closed-set identification, ultra-fine-grained temporal features necessary for open-set verification are inevitably compressed. Overall, these results strongly support the CLAE architecture as a highly efficient, compact speech representation model capable of extracting robust speaker and emotion information under severe resource constraints.

5 Conclusion

In this paper, we addressed the critical scarcity of efficient, language-specific speech models for Bengali by introducing a Continuous Latent Autoencoder (CLAE). While the current paradigm in low-resource speech AI heavily relies on fine-tuning massive, computationally expensive multilingual foundation models, we demonstrated that training a highly compact 23.8M-parameter model entirely from scratch is a highly viable and effective alternative. By leveraging a novel self-supervised learning objective that integrates mel-spectrogram reconstruction with a Joint-Embedding Predictive Architecture (JEPA) and Variance-Invariance-Covariance Regularization (VISReg), our model successfully captures rich, contextual phonetic nuances specific to Bengali.

Our evaluations indicate that this lightweight framework, paired with a small frozen feature extractor, not only generates robust representations at a 12.5 Hz latent sequence rate but also achieves performance that is competitive with, and in some metrics superior to, significantly larger state-of-the-art speech models. Ultimately, this work provides a computationally accessible, resource-friendly foundation for Bengali speech research and establishes a promising methodology for developing dedicated speech foundation models in other low-resource languages.

Limitations and Future Work

Our Continuous Latent Autoencoder (CLAE) offers a much more efficient alternative to large multilingual models, but there are a few limitations to keep in mind. First, we only trained the model on a limited set of publicly available 16 kHz Bengali speech datasets. Because of this, we haven't fully tested how well it holds up in really noisy, real-world environments or how it handles rare regional dialects that just aren't common in the training data. Also, while keeping the model small at 23.8M parameters is great for running it on basic hardware, it means the model misses out on the broader cross-lingual knowledge that massive foundation models pick up from hundreds of thousands of hours of audio. Finally, even though our model scored very well on certain specific metrics, we still have a way to go before it reaches the universal accuracy needed across all types of speech data to be used as a fully robust Automatic Speech Recognition (ASR) system in production.

Looking ahead, there is a lot of room to build on this work. Since our current experiments were limited by university hardware constraints and only used public datasets, the model still has plenty of untapped potential. One of the most promising takeaways was how smoothly the model converged during training, which points to really strong generalization capabilities. In the future, we or other researchers could scale up the CLAE architecture by adding more parameters and throwing much more training data at it to push its accuracy and generalization even higher for Bengali speech. An expanded version of this model could eventually power practical tools like Bengali voice typing and interactive chatbots. These kinds of tools are incredibly important for bridging the digital divide, helping millions of native Bengali speakers who don't know English finally get access to reliable speech AI technologies.

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References

Alexei Baevski, Yuhao Zhou, Abdelrahman Mohamed, and Michael Auli. 2020. wav2vec 2.0: A framework for self-supervised learning of speech representations. In *Advances in Neural Information Processing Systems*, volume 33, pages 12449–12460.

Sanyuan Chen, Chengyi Wang, Zhengyang Chen, Yu Wu, Shujie Liu, Zhuo Chen, Jinyu Li, Naoyuki Kanda, Takuya Yoshioka, et al. 2022. Wavlm: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1505–1518.

Antón de la Fuente and Dan Jurafsky. 2024. A layer-wise analysis of mandarin and english suprasegmentals in ssl speech models. *arXiv preprint arXiv:2408.13678*.

Ahana Deb, Sayan Nag, Ayan Mahapatra, et al. 2023. Beats: Bengali speech acts recognition using multimodal attention fusion. *arXiv preprint arXiv:2306.02680*.

Ratnajit Dhar and Arpita Mallik. 2026. 823-olt @ buet dl sprint 4.0: Context-aware windowing for asr and fine-tuned speaker diarization in bengali long form audio. *arXiv preprint arXiv:2602.21183*.

Tawsif Tashwar Dipto, Azmol Hossain, Rubayet Sabbir Faruque, et al. 2025. Are asr foundation models generalized enough to capture features of regional dialects for low-resource languages? *arXiv preprint arXiv:2510.23252*.

Wei-Ning Hsu, Benjamin Bolte, Yao-Hung Hubert Tsai, Kushal Lakhotia, Ruslan Salakhutdinov, and Abdelrahman Mohamed. 2021. Hubert: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29:3451–3460.

Vineel Pratap, Andros Tjandra, Bowen Shi, Paden Tomasello, Arun Babu, Sayani Kundu, Ali Elkahky, Zhaoheng Ni, Apoorv Vyas, et al. 2023. Scaling speech technology to 1,000+ languages. *arXiv preprint arXiv:2305.13516*.

Mohammed Rakib, Md. Ismail Hossain, Nabeel Mohammed, and Fuad Rahman. 2022. Bangla-wave: Improving bangla automatic speech recognition utilizing n-gram language models. *arXiv preprint arXiv:2209.12650*.

H. A. Z. Sameen Shahgir, Khondker Salman Sayeed, and Tanjeem Azwad Zaman. 2022. Applying wav2vec2 for speech recognition on bengali common voices dataset. *arXiv preprint arXiv:2209.06581*.

Ludovic Tuncay, Etienne Labbé, Emmanouil Benetos, and Thomas Pellegrini. 2025. Audio-jepa: Joint-embedding predictive architecture for audio representation learning. *arXiv preprint arXiv:2507.02915*.

Apoorv Vyas, Srikanth Madikeri, and Hervé Bourlard. 2021. Comparing ctc and lfmmi for out-of-domain adaptation of wav2vec 2.0 acoustic model. *arXiv preprint arXiv:2104.02558*.

Hadi Zaatiti, Hatem Hajri, Osama Abdullah, and Nader Masmoudi. 2025. Towards stable ai systems for evaluating arabic pronunciations. *arXiv preprint arXiv:2508.19587*.

Ruiying Zhu and Meng Shen. 2023. Fine-tuning convergence model in bengali speech recognition. *arXiv preprint arXiv:2311.04122*.

A Appendix

This is an appendix.